



Copilot

The threshold voltage (V_{th}) of a MOSFET is the minimum gate-to-source voltage (V_{gs}) required to turn the MOSFET 'ON', which allows current to start

working of *JFET*, it must be operated in the active region.

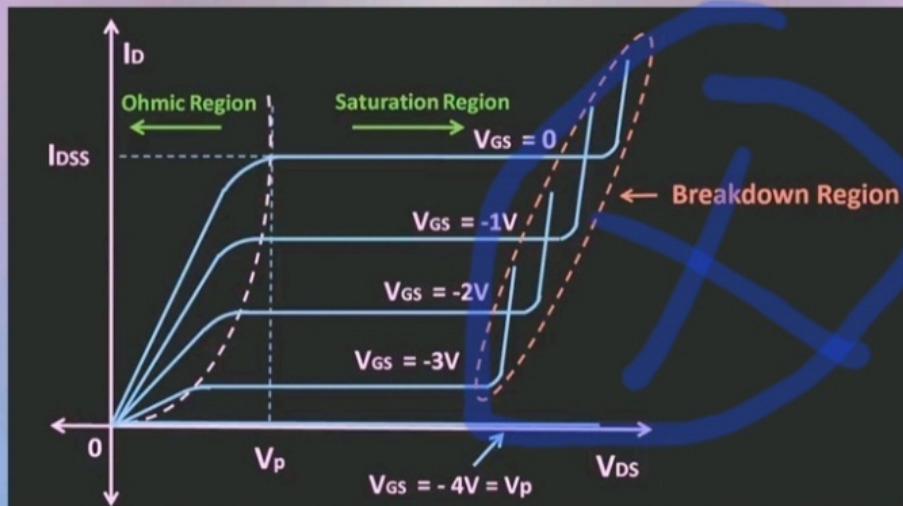
2. Pinch off Voltage (V_p). *It is the minimum drain-source voltage at which the drain current essentially becomes constant.*

Figure 19.11 shows the drain curves of a *JFET*. Note that pinch off voltage is V_p . The highest curve is for $V_{GS} = 0V$, the shorted-gate condition. For values of V_{DS} greater than V_p , the drain current is almost constant. It is because when V_{DS} equals V_p , the channel is effectively closed and does not allow further increase in drain current. It may be noted that for proper function of *JFET*, it is always operated for $V_{DS} > V_p$. However, V_{DS} should not exceed $V_{DS(max)}$ otherwise *JFET* may breakdown.

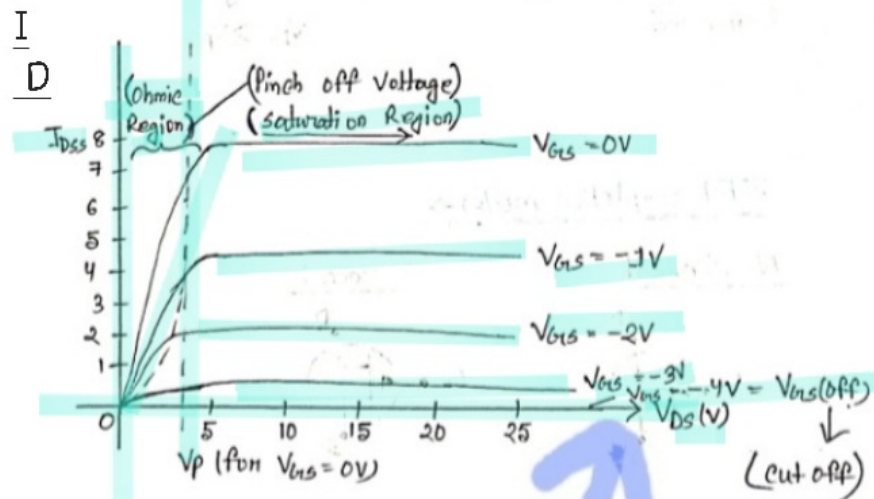
I_D (mA)

I_D

Drain or Output Characteristics Curve of JFET



I-V characteristic curve:



It is the curve between drain current (I_D) and drain source voltage (V_{DS}) for different gate source voltage (V_{GS}). It can be characterized as:

1. For $V_{GS} = 0V$ the drain current is maximum. It is denoted as I_{DSS} and called shorted gate or drain current.
2. If V_{GS} increases drain current I_D decreases ($I_D < I_{DSS}$) even though V_{DS} is increased.
3. When V_{GS} reaches a certain value, the drain current will be decreased to zero.
4. For different V_{GS} , the I_D will become constant after pinch off voltage (V_p) though V_{DS} is increased.

Feature	BJT	JFET
Operating Principle	Current-controlled	Voltage-controlled
Construction	3 doped layers	Single doped region
Input Impedance	Lower	Higher
Gain	Generally higher	Generally lower
Noise Level	Higher	Lower
Applications	Amplifiers, switches	High-impedance circuits, low-noise amplifiers



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An electronic circuit that generates square waves (or other non-sinusoidal waves such as rectangular, saw-tooth waves) is known as a ***multivibrator**.

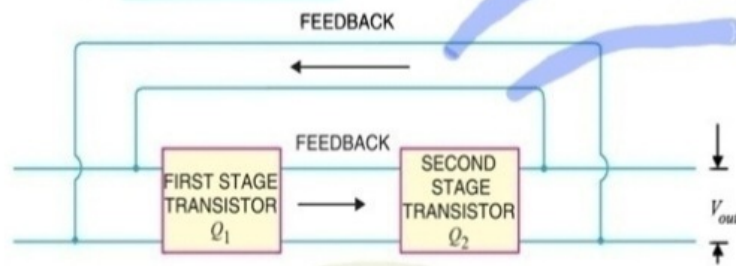


Fig. 18.10

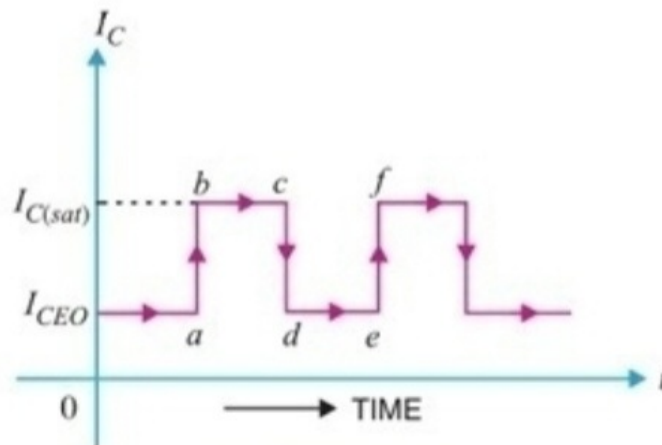


Fig. 18.11

The circuit, a two-stage amplifier, operates in two states: ON and OFF. These states are controlled by the circuit conditions and feedback from each amplifier stage. One stage is driven to saturation (ON), while the other is cut off (OFF). After a certain time, the states reverse.

The output, which can be a rectangular or square wave, is taken from either stage. The multivibrator, shown in Fig. 18.10, has 100% positive feedback. At any moment, one transistor is ON and the other is OFF. If Q_2 is ON and Q_1 is OFF, the collector current in Q_2 will be $I_{C(sat)}$. This state lasts for a time (bc in this case) set by the circuit conditions.

Then, Q_2 is cut off and Q_1 is turned ON. The collector current in Q_2 is now I_{CEO} . The circuit remains in this state for a time de. The cycle repeats, producing a square wave output.

Example 25.33. For the noninverting amplifier circuit shown in Fig. 25.58, find the output voltage for an input voltage of (i) 1 V (ii) -1 V.

Solution. Voltage gain, $A_{CL} = 1 + \frac{R_f}{R_i} = 1 + \frac{10 \text{ k}\Omega}{1 \text{ k}\Omega} = 1 + 10 = 11$

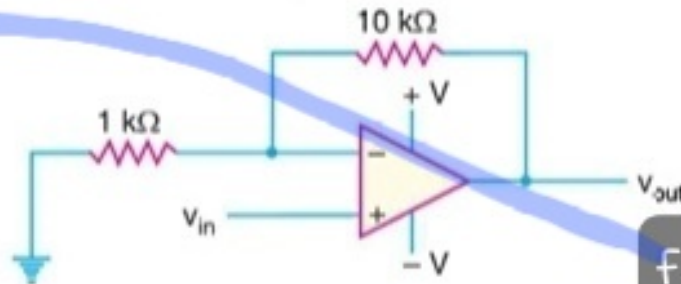


Fig. 25.58

(i) For $v_{in} = 1 \text{ V}$; $v_{out} = A_{CL} \times v_{in} = 11 \times 1 \text{ V} = 11 \text{ V}$

(ii) For $v_{in} = -1 \text{ V}$; $v_{out} = A_{CL} \times v_{in} = 11 \times (-1 \text{ V}) = -11 \text{ V}$

Example 25.34. For the noninverting amplifier circuit shown in Fig. 25.59, find peak-to-peak output voltage.

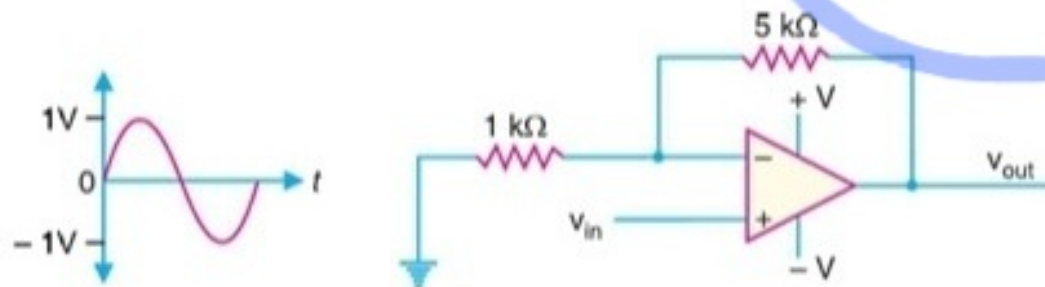
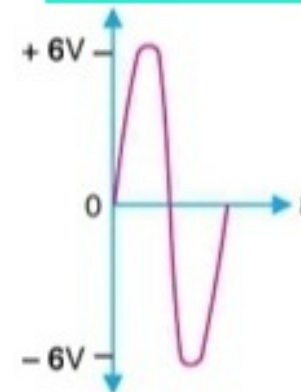


Fig. 25.59

Solution. The input signal is 2 V peak-to-peak.

Voltage gain, $A_{CL} = 1 + \frac{R_f}{R_i} = 1 + \frac{5 \text{ k}\Omega}{1 \text{ k}\Omega} = 1 + 5 = 6$

$\therefore$ Peak-to-peak output voltage $= A_{CL} \times v_{inpp} = 6 \times 2 = 12 \text{ V}$



Positive feedback is generally not used to cancel noise in a system. This is because positive feedback tends to amplify the signal, and if noise is present, it will also be amplified, which is undesirable.

Negative feedback, on the other hand, is commonly used for noise cancellation. Negative feedback works by taking a portion of the output signal and feeding it back to the input in a manner that cancels out the noise. This results in a cleaner output signal.

Now, let's derive the negative voltage feedback gain (A_{vf}) of an operational amplifier.

Consider an operational amplifier (op-amp) with negative feedback where:

- A_v is the open-loop gain of the op-amp
- β is the feedback factor
- V_{in} is the input voltage
- V_{out} is the output voltage

The gain of the op-amp with negative feedback (A_{vf}) is given by:

$$A_{vf} = \frac{V_{out}}{V_{in}}$$

Without feedback, the output of the op-amp is:

$$V_{out} = A_v \cdot V_{in}$$

due to feedback v_{in}

With negative feedback, a portion of the output (βV_{out}) is fed back to the input. Therefore, the effective input voltage becomes ($V_{in} - \beta V_{out}$). So, the output becomes:

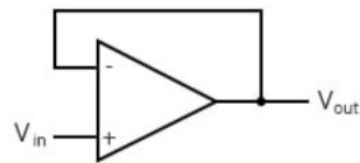
$$V_{out} = A_v \cdot (V_{in} - \beta V_{out})$$

$$V_{out}(1 + A_v\beta) = A_v \cdot V_{in}$$

Therefore, the gain with feedback becomes:

$$A_{vf} = \frac{V_{out}}{V_{in}} = \frac{A_v}{1 + A_v\beta}$$

As you can see, the gain is reduced compared to the open-loop gain due to the presence of the feedback term in the denominator. This reduction in gain helps to improve the stability of the system and reduce the impact of noise.



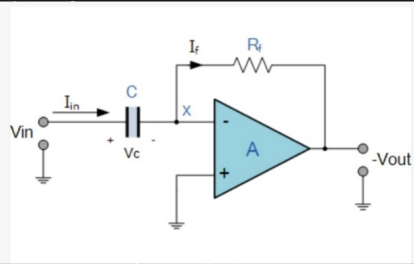
Negative Feedback | Operational Amplifiers | Electronics Textbook

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An op-amp works in a differentiator amplifier by producing an output voltage that changes as fast as the input voltage does. If the input voltage changes quickly, the output voltage is high. If the input changes slowly or not at all, the output voltage is low or zero. It's like the op-amp is watching how fast the input voltage is moving and adjusting its output to match that speed.

You

Differentiator amplifier configuration works using an op-amp. Derive the formula for calculating the gain in



Here's how you can derive the formula for the voltage gain (A_v) in a differentiator amplifier:

1. **Input Impedance:** The input impedance of the capacitor (Z_C) is given by:

$$Z_C = \frac{1}{j\omega C}$$

where:

- (j) is the imaginary unit.
- (ω) is the angular frequency of the input signal.
- (C) is the capacitance.

2. **Voltage Gain:** The voltage gain (A_v) of the amplifier is the ratio of the output voltage (V_{out}) to the input voltage (V_{in}):

$$A_v = \frac{V_{out}}{V_{in}}$$

3. **Output Voltage:** The output voltage (V_{out}) is the product of the input current (I_{in}) and the feedback resistor (R_f):

$$V_{out} = -I_{in} \cdot R_f$$

4. **Input Current:** The input current (I_{in}) is the voltage across the capacitor (V_{in}) divided by the input impedance (Z_C):

$$I_{in} = \frac{V_{in}}{Z_C}$$

5. **Substitute (I_{in}) in (V_{out}):**

$$V_{out} = -\frac{V_{in}}{Z_C} \cdot R_f$$

6. **Substitute (Z_C) in (V_{out}):**

$$V_{out} = -V_{in} \cdot j\omega C R_f$$

7. **Final Gain Formula:** Now, substituting (V_{out}) in the gain formula (A_v), we get:

$$A_v = \frac{-V_{in} \cdot j\omega C R_f}{V_{in}}$$

$$A_v = -j\omega C R_f$$

Therefore, the gain of a differentiator amplifier is proportional to the frequency of the input signal, the capacitance, and the feedback resistance. The

$$= -(100 \times 10^3 \Omega) \times (10 \times 10^{-6} \text{ F}) \frac{d}{dt} = -\frac{1}{dt}$$

(ii) Since the input voltage is a straight line between 0 and 0.2s, the output voltage is

$$v_o = -\frac{dv_i}{dt} = -\frac{(1-0)}{0.2} = -5\text{V}$$

Therefore, between 0 to 0.2s, the output voltage is constant at -5 V . For $t > 0.2\text{ s}$, the input is constant so that output voltage is zero. Fig. 25.92 (ii) shows the output waveform.

25.38. Comparators

Often we want to compare one voltage to another to see which is larger. In this situation, a *comparator* may be used. A *comparator* is an *OP-amp* circuit without negative feedback and takes advantage of very high open-loop voltage gain of *OP-amp*. A comparator has two input voltages (noninverting and inverting) and one output voltage. Because of the high open-loop voltage gain of an *OP-amp*, a very small difference voltage between the two inputs drives the amplifier to saturation. For example, consider an *OP-amp* having $A_{OL} = 100,000$. A voltage difference of only 0.25 mV between the inputs will produce an output voltage of $(0.25\text{ mV})(100,000) = 25\text{ V}$. However, most of *OP-amps* have output voltages of less than $\pm 15\text{ V}$ because of their d.c. supply voltages. Therefore, a very small differential input voltage will drive the *OP-amp* to saturation. This is the key point in the working of comparator.

Fig. 25.93 illustrates the action of a comparator. The input voltages are v_1 (signal) and v_2 (*reference voltage). If the differential input is positive, the circuit is driven to saturation and output goes to maximum positive value ($+I'_{sat} = +13\text{ V}$). Reverse happens when the differential input goes negative *i.e.* now output is maximum negative ($-I'_{sat} = -13\text{ V}$). This circuit is called comparator because it compares v_1 to v_2 to produce a saturated positive or negative output voltage. Note that output voltage rapidly changes from -13 V to $+13\text{ V}$ and *vice-versa*.

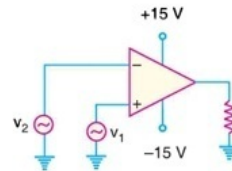


Fig. 25.93

25.39 Comparator Circuits

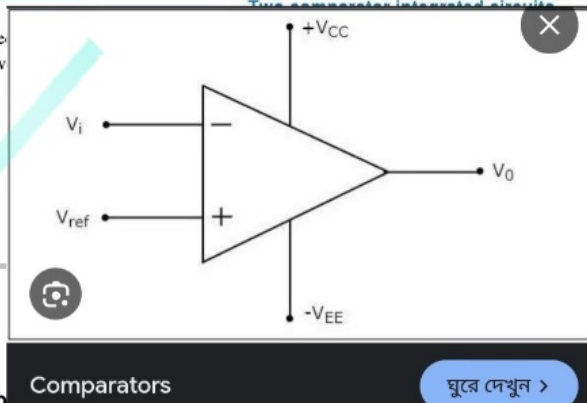
A comparator circuit has the following two characteristics :

- It uses no feedback so that the voltage gain is equal to the open-loop voltage gain (A_{OL}) of *OP-amp*.
- It is operated in a non-linear mode.

These properties of a comparator permit it to perform many useful functions.



- * If this terminal is grounded
- ** Since in our case supply v



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Comparators

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1. As a square wave generator. A comparator can be used to produce a square wave output from a sine wave input. Fig. 25.94 shows the circuit of a comparator to produce square wave output. Note that inverting terminal ($-$) is grounded and signal (v_{in}) is applied to the noninverting terminal ($+$). Since the gain of a comparator is equal to A_{OL} , virtually any difference voltage at the inputs will cause the output to go to one of the voltage extremes ($+I'_{sat}$ or $-I'_{sat}$) and stay there until the voltage difference is removed. The polarity of the input difference voltage will determine to which extreme ($+I'_{sat}$ or $-I'_{sat}$) the output of the comparator goes.

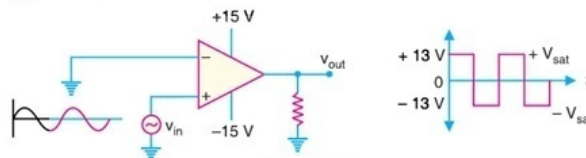


Fig. 25.94

When the input signal goes positive, the output jumps to about $+13\text{ V}$. When the input goes negative, the output jumps to about -13 V . The output changes rapidly from -13 V to $+13\text{ V}$ and *vice-versa*. This change is so rapid that we get a square wave output for a sine wave input.

2. As a zero-crossing detector. When one input of a comparator is connected to ground, it is known as zero-crossing detector because the output changes when the input crosses 0 V . The zero-crossing circuit is shown in Fig. 25.95. The input and output waveforms are also shown. When the input signal is positive-going, the output is driven to positive maximum value (*i.e.* $+I'_{sat} = +13\text{ V}$). When the input crosses the zero axis and begins to go negative, the output is driven to negative maximum value (*i.e.* $-I'_{sat} = -13\text{ V}$).

